



AI Tool Prompt Engineering Report

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AI Tool Overview

The purpose of this paper is to examine the differences between self-made prompts and AI-enhanced prompts on the output of one AI tool of choice. The AI tool used in this report was NotebookLM, which tailors its responses to the user's uploaded material. In the AI Tool Graph, it falls under the Scientific and Academic Research domain, specifically Source-Ground Analysis. It is used to synthesize and draw insights from specific materials and sources, which improve workflow, learning, and research (Google Notebook LM, n.d.). I chose this tool since it can be useful in note-taking, studying, and drawing insights for research. In marketing, research serves as a valuable tool in tailoring marketing strategies and campaigns to the needs of the target market. Therefore, it is relevant to my field of study.

Original Prompt

Before entering my prompt, I uploaded a PDF of a material of my choosing. The PDF contained a chapter of the e-book material currently used in a subject that I am taking. My original prompt after uploading the PDF material was: *"Give me a summarization of the entire chapter using the source given."* The PDF material and a screenshot of the prompt are presented below.

Uploaded PDF Material:

 Consumer Behavior Chapter 7.pdf

Wednesday, March 18

Give me a summarization of the entire chapter using the source given

A zoomed-in photo of my original prompt

The resulting response by the AI was as follows:

The screenshot shows the NotebookLM interface with the title "Consumer Attitudes and the Hierarchy of Persuasion". The "Sources" panel on the left lists "Consumer Behavior Chapter 7.pdf". The "Chat" panel in the center contains the following text:

Wednesday, March 18

Give me a summarization of the entire chapter using the source given

Chapter 7 explores the complex nature of consumer attitudes, how they are formed, and the various strategies marketers use to change them.

Understanding Attitudes

Attitudes are defined as relatively enduring overall evaluations of products, services, issues, or people. 1. According to the **ABC approach**, attitudes consist of three components: **affect** (feelings), **behavior** (intentions or overt actions), and **cognitions** (beliefs). 2. These attitudes serve four primary functions:

- **Utilitarian:** Used to obtain rewards or avoid punishment. 3 4.
- **Knowledge:** Allows consumers to simplify decision-making. 3 4.
- **Value-expressive:** Enables consumers to express their core values and self-concept. 3 4.
- **Ego-defensive:** Acts as a defense mechanism to protect self-image or avoid threatening information. 4 5.

The Hierarchy of Effects

Research suggests that the three components of attitude (ABC) form in a sequential pattern known as the **hierarchy of effects**. 6. The order depends on the purchase context:

- **High-involvement:** Beliefs → Affect → Behavior. 7.
- **Low-involvement:** Beliefs → Behavior → Affect. 7.
- **Experiential:** Affect → Behavior → Beliefs. 7 8.

Start typing...

1 source

The "Studio" panel on the right shows a grid of tool icons: Audio Overview, Side Deck (BETA), Video Overview, Mind Map, Reports, Flashcards, Quiz, Info (BETA), and Data Table. Below the grid is a note titled "The Architecture of Consumer Attitudes and..." with a "1d ago" timestamp and an "Add note" button.

The screenshot shows the NotebookLM interface with the same title "Consumer Attitudes and the Hierarchy of Persuasion". The "Sources" panel remains the same. The "Chat" panel now contains the following text:

Behavioral influence: Behavior → Beliefs → Affect. 7 8.

Consumer Attitude Models

The sources highlight two major models for measuring attitudes:

1. **Attitude-Toward-the-Object (ATO) Model:** Also known as the Fishbein model, it considers salient beliefs about attributes, the strength of those beliefs, and an evaluation of the attributes. 9 10. It is a **compensatory model**, meaning poor ratings on one attribute can be offset by high ratings on another. 11.
2. **Behavioral Intentions Model:** An improvement on the ATO model that focuses on intentions to act, incorporates subjective norms (perceptions of what others think), and focuses on the behavior of buying rather than just the object itself. 12 13.

Persuasion and Attitude Change

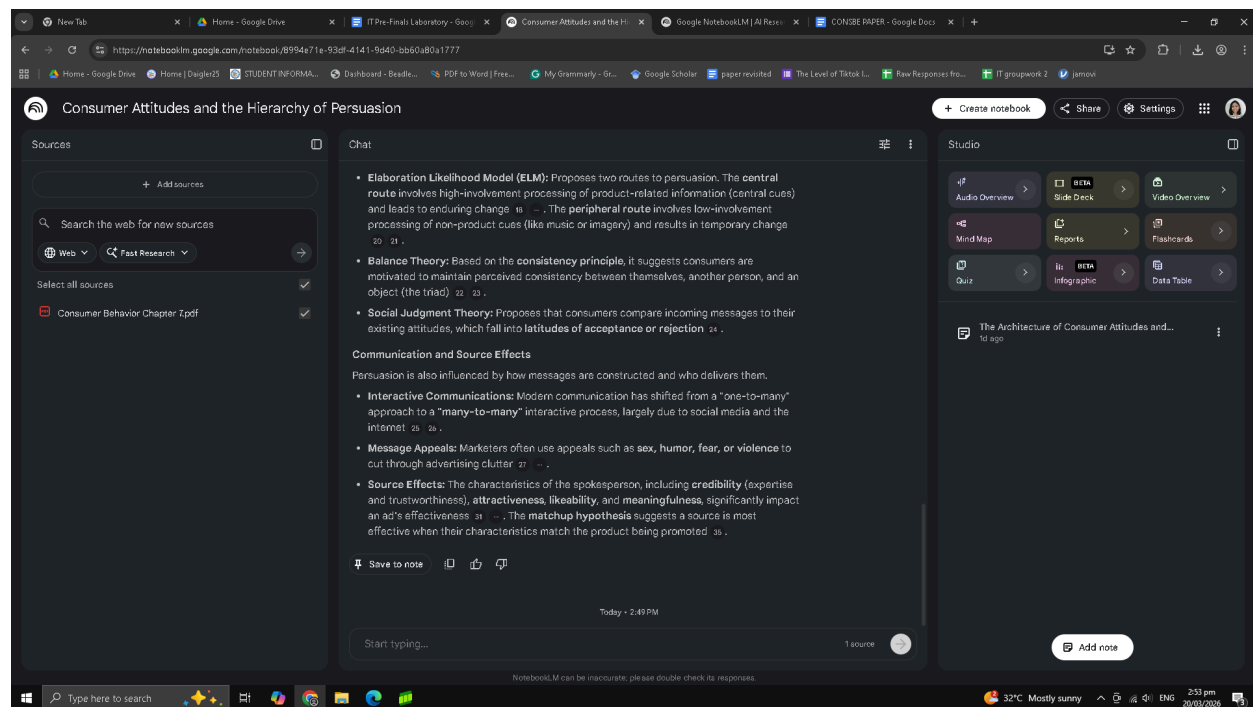
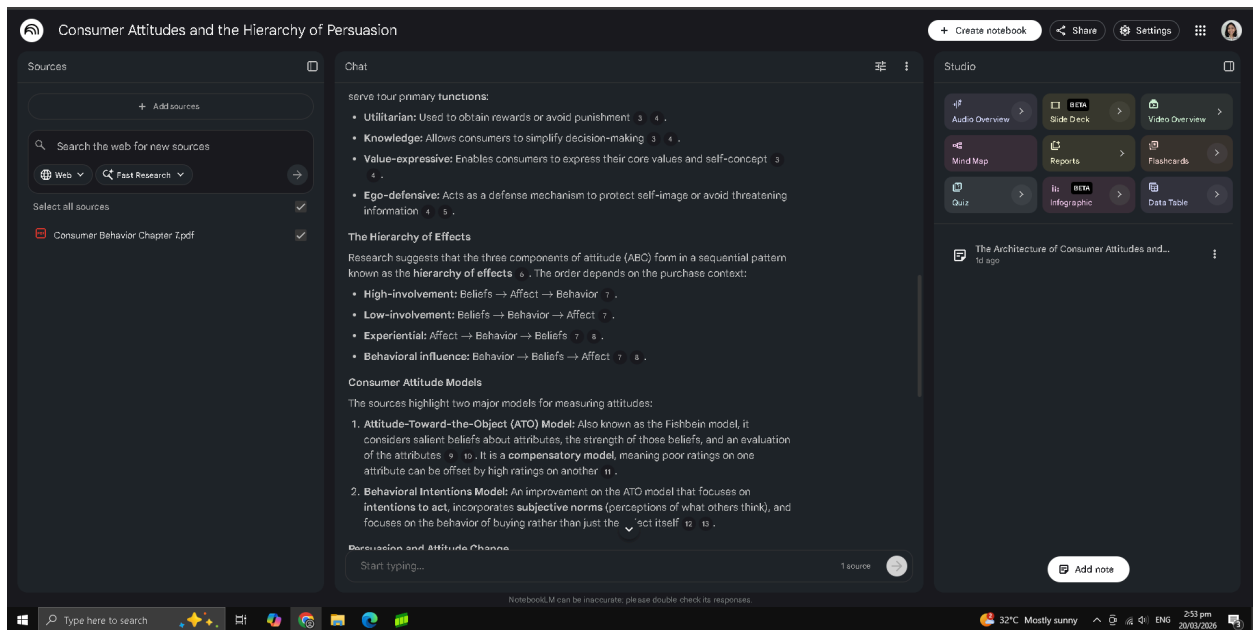
Persuasion refers to specific attempts to change attitudes. 14. Several theories explain this process:

- **ATO Approach:** Marketers can change beliefs, add new beliefs about attributes, or change evaluations of existing attributes. 15 16.
- **Elaboration Likelihood Model (ELM):** Proposes two routes to persuasion. The **central route** involves high-involvement processing of product-related information (central cues) and leads to enduring change. 18 19. The **peripheral route** involves low-involvement processing of non-product cues (like music or imagery) and results in temporary change. 20 21.
- **Balance Theory:** Based on the **consistency principle**, it suggests consumers are motivated to maintain perceived consistency between themselves, another person, and an object (the triad). 22 23.
- **Social Judgment Theory:** Proposes that consumers compare incoming messages to their existing attitudes, which fall into latitudes of acceptance or rejection. 24.

Start typing...

1 source

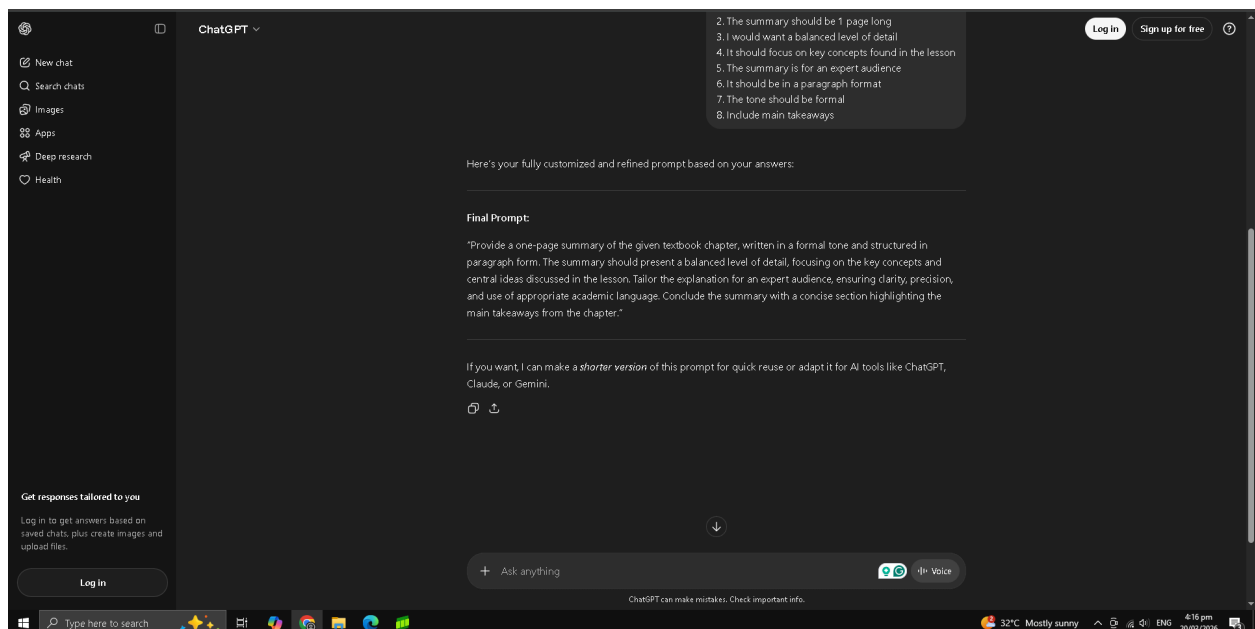
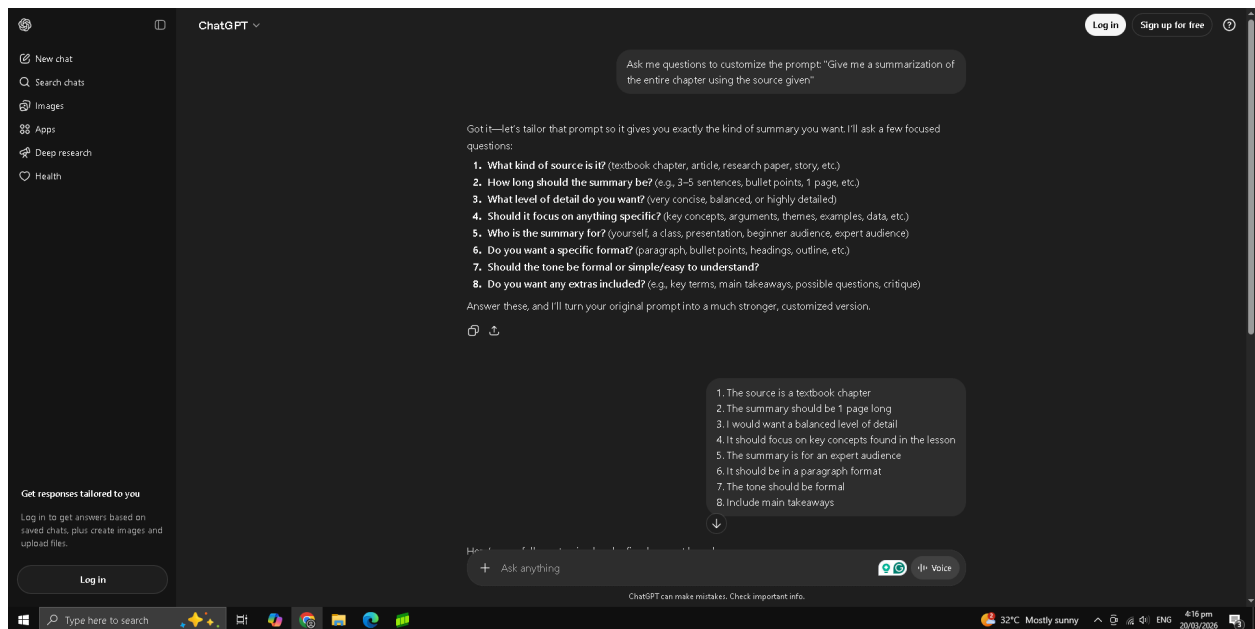
The "Studio" panel on the right is identical to the first screenshot, showing the same grid of tool icons and the note titled "The Architecture of Consumer Attitudes and...".



The results of the initial prompt

AI-Assisted Prompt

Based on the initial prompt results, I enhanced it with the Large Language Model (LLM) ChatGPT. To do this, I pasted my previous prompt and asked the LLM to help me customize and improve it. It then asked me 8 questions about how I wanted the output to be. Below are screenshots detailing the process of improving the original prompt:



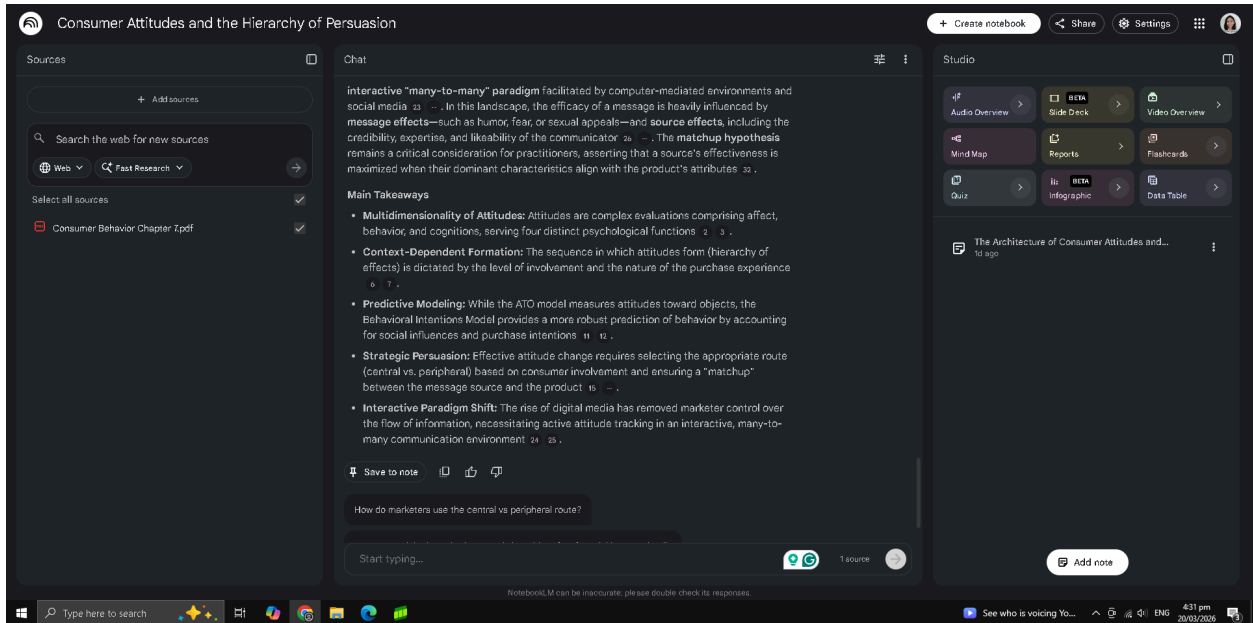
Process of improving the initial prompt through ChatGPT

The new prompt will then be: *“Provide a one-page summary of the given textbook chapter, written in a formal tone and structured in paragraph form. The summary should present a balanced level of detail, focusing on the key concepts and central ideas discussed in the lesson. Tailor the explanation for an expert audience, ensuring clarity, precision, and use of appropriate academic language. Conclude the summary with a concise section highlighting the main takeaways from the chapter.”*

Going back to NotebookLM, I input the new prompt, which resulted in the following:

The screenshot shows the NotebookLM interface with a notebook titled "Consumer Attitudes and the Hierarchy of Persuasion". The chat window displays a prompt: "Provide a one-page summary of the given textbook chapter, written in a formal tone and structured in paragraph form. The summary should present a balanced level of detail, focusing on the key concepts and central ideas discussed in the lesson. Tailor the explanation for an expert audience, ensuring clarity, precision, and use of appropriate academic language. Conclude the summary with a concise section highlighting the main takeaways from the chapter." The response is a detailed summary of consumer attitudes and the hierarchy of effects, discussing concepts like the ABC approach, functional theory of attitudes, and the Attitude-Toward-the-Object (ATO) model. The interface includes a Sources panel on the left, a Chat panel in the center, and a Studio panel on the right with various tools like Audio Overview, Slide Deck, and Video Overview.

This screenshot shows the same NotebookLM interface, but the chat window displays a different response. The response focuses on the Object (ATO) model, the Elaboration Likelihood Model (ELM), and the Matchup Hypothesis. It discusses how persuasion is explained through theoretical lenses like the ATO model and ELM, and how modern marketing communication has transitioned from a linear "one-to-many" model to an interactive "many-to-many" paradigm. The response concludes with a section on Main Takeaways, highlighting the multidimensionality of attitudes and the context-dependent formation of attitudes. The interface elements (Sources, Chat, Studio) are consistent with the previous screenshot.



The results of the AI-assisted prompt

Comparison and Analysis

The main difference that I observed between the two outputs was their formatting. I noticed that the output from the initial prompt only gave a brief description of the chapter's contents. However, with the AI-assisted prompt, the output was closer to what I wanted. For me, the output generated with the AI-assisted prompt was more useful, as it was specific, formal, and suitable for submission.

The improved prompt from ChatGPT affected the clarity, depth, and relevance of the output generated by NotebookLM by being more detailed. First, it added a sentence in the prompt, stating that the output should be tailored to an educated, expert audience. Moreover, ChatGPT added a sentence. In that way, the result was clear and understandable. Second, because the new prompt specified the level of detail to be included in the output, the output became deeper, rather than a list of keywords and their definitions. Lastly, it strengthened the relevance of the output through its delivery and accuracy to the source material.

The improvement in the generated output lies in the prompt. Specifically, the addition of formatting and tone made the output more expansive than just a short description of the key terms in the chapter.

Overall, this was a meaningful experience for me because it showed how much an AI's output can change based on the prompt. I learned that in prompt engineering, short prompts do not necessarily give the best results in terms of detail and content. However, they are useful for

other purposes, such as brainstorming and a quick understanding of certain topics. On the other hand, longer prompts allow for an exhaustive discussion of the source given, from which one can gain insights. Ultimately, it depends on the person using the AI tool what they want their output to be: an essay, a summary, or a list of key concepts and terms.

Reference

Google Notebook LM. (n.d.). <https://sites.google.com/view/notebook-lm>